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Influence of emojis on user engagement in brand-related user generated content

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ABSTRACT

Emojis are increasingly adopted in various platforms, such as text messages, social media, or blogs, as part of other digital communications. Recently, emojis have been used in brand-related user-generated content (UGC) as a strategic communication tool to promote positive outcomes. In spite of their increasing importance, little is known about the impact of incorporating emojis in brand-related UGC on consumer reactions. In particular, few research studies have investigated the joint effect of emojis with texts, which is the most important context where emojis function, and the contextual conditions that affect the influence of emojis on consumer reactions. To fill this research gap, we study brand-related UGC, focusing on the effect of emojis and the contextual conditions (e. g., texts). Using large-scale brand-related social media posts from Instagram, we find that, overall, the presence of emojis is positively associated with consumer engagement. Economically, the presence of emojis increases the average number of likes by 72% and the average number of comments by 70% in comparison to posts without any emojis. The interaction effects between emojis and texts reveal that emotional emojis have a positive and significant relationship with consumer engagement only when the texts in brand-related UGC are being skewed toward positive sentiments. However, informational emojis are negatively related with consumer engagement in the similar context. Additionally, the investigations regarding the contextual conditions show that using more emotional emojis has a positive influence on consumer engagement in commercial posts but a negative influence in general posts.

1. Introduction

Emojis are an emerging communication tool, the use of which has increased on social media sites in recent years. Strategic social media communicators employ emojis for various purposes, such as adding extra context to their content or helping audiences better understand their messages, expecting positive outcomes. While emojis are generally expected to function as a content promoter[1] (Molina et al., 2019; Riordan, 2017), the literature regarding the influences of emojis suggests mixed effects. The effects may either be contextually dependent (Thollander & Kumar, 2019; Lu et al., 2016) or vary across different types of emojis (Hewage et al., 2020; Barbieri et al., 2016). In particular, research regarding emojis' interactions with texts is still limited in spite of the fact that, in many cases, emojis function in a way that complements texts. There are some occasions where emojis are used without any words, but they are usually understood in context. For instance, in social media, emojis help content creators add tone and clarity to their communications with the post viewers.

Another focus of this study is user engagement in UGC of a social media platform. Because computer users participate in social conversations as content creators or audiences and actively spread the content using their dynamic networks, UGC has become an increasingly important tool to influence users (Castillo et al., 2021; Hartmann et al., 2021; Ding et al., 2017; Goh et al., 2013; Zeng & Wei, 2013; Ye et al., 2011). For example, consumers' engagement with brands in social media posts is a recent trend related to UGC. Using easy access to

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[1] According to WordStream (2018), an emoji in a Tweet increases user engagement by 25%, and an emoji in Facebook posts can make the post more viral by increasing shares by 33% and interactions by 57% (Zazzle Media). https://www.wordstream.com/blog/ws/2015/11/19/twitter-emoji https://www.zazzlemedia.co.uk/blog/facebook-edgerank-and-engagement/.

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advances in recent technology and dynamic online networks, ordinary consumers create an extensive amount of brand-related UGC, a content that contains brand names as hashtags (e.g., *#nike*), and they share their experiences with brands and products. The vast amount of content, combined with the viral nature of the social communication tools, has significant potential business value for brands or products.

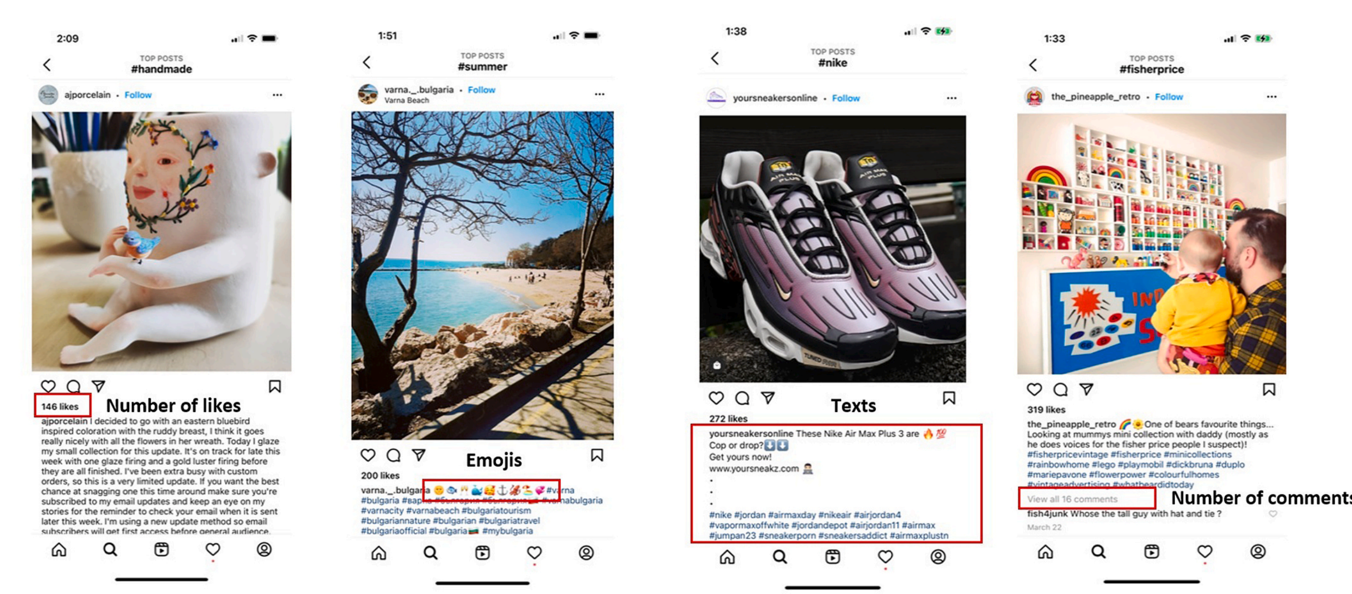
In this study, we view brand-related UGC as an important communication source and examine how emojis are effectively employed in brand-related social media posts to make social conversations more engaging or how they enrich texts by productively complementing texts, focusing on the impact of the emojis' properties on user engagement and their effective coupling methods with texts in social media. We also aim to determine what significant contextual conditions social media content creators need to consider by exploring the internal or external factors that may affect the relationship between emojis and user engagement.

Fig. 1 exhibits brand-related UGC in Instagram. Fig. 1(a) and (b) display a post without emojis and a post with emojis, respectively. Fig. 1 (c) and (d) present a post published by an Instagram user with a commercial purpose and a post by an ordinary user, respectively. Except for Fig. 1(a), the other three representative posts contain emojis along with texts. We posit that emojis are associated with consumer engagement measured by the number of likes and comments. We study the main effect of emojis, the interaction effect between the emojis and texts, and other contextual conditions that may impact the influence of emojis on consumer engagement, asking following research questions:

1) Is having emojis in brand-related social media posts positively associated with user engagement?
2) Is the influence of emojis on consumer engagement contingent on text properties in brand-related social media posts?
3) Does the influence of emojis on consumer engagement vary across different types of posts?

To examine these research questions, we leverage an observational dataset from Instagram. Instagram has been an increasingly popular social networking service since it was launched in October 2010; it is used as a social conversation tool by ordinary social media users and by brands. Since our focus is brand-related UGC in this study, we extract UGC using brand names as hashtags and delete the accounts managed by brands or corporates (i.e., brand accounts) while we are preprocessing the extracted raw data. Although our data primarily capture brand-related UGC published by general social media users after the business-owned accounts are removed, they still contain the posts that are associated with the accounts whose main purpose is to promote a retail business. These are not official Instagram business accounts (e.g., brand accounts); however, their main purpose is commercial-related, and they are run by small stores or individuals for business purposes. Unlike brand accounts, we include the data generated by small stores or individuals with commercial purposes in our analyses and label them as commercial posts. The commercial (versus general) posts are labeled based on user profiles; if a user profile contains the word that reflects the user's retailing-related purposes, the associated post is labeled as a commercial post. The user-generated social media post that is not a commercial post is defined as a general post. We further elaborate the operationalization and the labeling procedure in Section 3 (3.2.3).

Our empirical analyses provide several key findings. First, having emojis is likely to increase consumer engagement in brand-related UGC, but a greater use of emojis may not be beneficial. The economic interpretation of these findings would suggest that the number of likes (comments) increases by 72% (70%) in brand-related UGC with emojis in comparison to the UGC without any emojis. Second, while the main effect of emotional emojis on consumer engagement is trivial, its interaction effects with texts are noteworthy. We find that there is a synergetic relationship between emotional emojis and text valence. Emotional emojis are likely to increase consumer engagement in brand-related UGC, but only in the posts containing texts that are being skewed toward positive valence. In contrast, informational emojis are likely to decrease consumer engagement in the similar social media posts, the posts with positive texts. Finally, we find that the influence of emojis on consumer engagement is contingent on the type of a post. The analyses reveal that emotional emojis are positively associated with consumer engagement but only in commercial posts.



(a) A post without emojis (b) A post with emojis (c) A commercial post (d) A general post

Fig. 1. *Examples of Brand-related UGC.*

2. Research background and hypotheses development

2.1. Research background

We review the relevant literature on which we base and advance our research questions. Overall, the current research is related to two fields of literature.

The first is the effect of emojis or the effect of emojis in relation to texts on consumer reactions. The recent proliferation of emojis and the positive effects in persuasive communications have motivated social media content creators to integrate them into their social conversations. In response to the popularity and significant impact of emojis, scholars have studied on how consumers react to them in diverse contexts (e.g., emails, social media). In many cases, emojis are used alongside texts, and they function in a way that complement or supplement texts. For example, Internet users employ emojis to supplement texts and provide additional details regarding objects or events or to express their emotions. Extant research has documented the direct impacts of emojis on consumer reactions or how emojis in combination with texts play a role in consumer reactions. Some research has shown that employing emojis in messages may positively influence purchase intentions, and consumers' affective states account for the mechanism behind the result (Coyle & Carmichael, 2019; Das et al., 2019; Li et al., 2019; Molina et al., 2019). For example, Das et al. (2019) show that emojis in advertising communications lead to consumers' higher purchase intentions through their positive affect. Cramer et al. (2016) study the motivations of selecting certain emojis and their intended functions in one-on-one messaging (e.g., text, chat, or app). Unlike many other papers that connect emojis with consumer reactions that utilize surveys or experiments, Molina et al. (2019) use observational data to study the relationship between emojis and user engagement.

Although emojis are frequently coupled with texts in various communications, such as emails, chats, or social media, little research exists on how emojis are associated with consumers' reactions when used jointly with texts. Only a couple of papers document the joint effect. For example, Ai et al. (2017) show that emojis play both complementary and supplementary roles for texts. Specifically, they find that emotion-related emojis play a complementary role in relation to texts while entity-related emojis play a supplementary role in a message as alternatives to words. Different types of emojis and their impact on consumer reactions are also an under-studied area, but a few studies have documented interesting findings. Hu et al. (2017) study the intentions of using emojis and the interaction effect between emoji sentiment and message sentiment. They find that positive (negative) emojis are associated with an increase (decrease) in the sentiment of a plain positive (negative) verbal message. Hewage et al. (2020) compare asymmetric facial expression emojis and their symmetric counterparts and reveal that asymmetric emojis are more likely to receive favorable consumer evaluations. While most previous research on the effect of emojis focus on its global impact on consumer reactions or examine emoji types or text effect separately, emojis do not function in isolation. In most cases, emojis are interpreted within context, and they add to or modify the tone of the texts that they accompany. Thus, it is important to understand emojis in conjunction with texts, and how emojis and texts jointly influence consumers. In this paper, we extend the existing literature of consumer engagement prediction based on emojis and focus on how emojis and texts jointly influence consumer engagement. Some of the related literature examine the usages or semantics of emojis within messages (Ai et al., 2017; Hill, 2016; Kaye et al., 2021), but our study differs from the existing research in that we differentiate emojis by segmenting them into emotional and informational emojis, and examine how the different types of emojis play a role in conjunction with texts. We also examine more diverse dimensions of texts in which the different types of emojis may function and impact consumer responses.

The second field of literature that is related to our study is the contextual conditions that may impact the influence of emojis on consumer reactions, which is closely related to another focus of this research: examining the particular contexts that may explain when or where emojis are more or less influential in brand-related UGC. Extant research has focused on the settings where emojis are used, the demographic or cultural differences that account for the varied emoji effects, or the product types upon which the emoji impact is contingent (Jones et al., 2020; Das et al., 2019; Lu et al. 2016). Das et al. (2019) examine the factors that impact the emoji effect on consumers' purchase intentions; they find that the positive emoji effect depends on the product type (utilitarian vs. hedonic). Their experiments show that emojis are only effective for hedonic products, not for utilitarian products. Several studies investigate the usages or interpretations of emojis across different cultures or languages. Lu et al. (2016) explore cultural differences on emoji preferences and find that the categories and frequencies of emojis are important signals for the cultural variations. A study by Thollander and Kumar (2019) investigates how the adoption and perception of emojis vary across different cultural backgrounds. Surveys and interviews fill the gap between user perceptions and the current emoji standard. With 30 million tweets, Barbieri et al. (2016) conduct a research study on the semantics of emojis; they find that the top 150 most popular emojis semantically concur across languages while some emojis are used in different contexts across distinct languages. Using emojis in professional communications may not be beneficial. A study by Haberstroh (2010) focus on the effect of informal language, such as emoticons in professional settings, and reveal that using emoticons following the clients' formal expressions is detrimental for their perceptions of a counselor's expertise. Chen et al. (2018) argue that emoji usage is contingent on gender (females vs. males). Using a machine learning algorithm, they provide the evidence that the gender of a user can be detected by the emoji usage in their messages. Li et al. (2019) examine how emoticons are perceived by consumers in commercial relationships; they find that the effect is contingent upon customer types (communal-oriented vs. exchange-oriented). While existing research on the moderating factors that account for the contingency of emoji impact has explained something interesting, these settings oversimplify the contexts (where or when) in which the emojis are employed. In this study, we extend the existing settings where emojis efficiently function by incorporating a richer set of conditional factors and by observing the contexts that promote or dissuade emoji usage.

Overall, we aim to fill the research gap and test our research questions empirically with large-scale observational data. There is extant research that studies emojis, but we delve further into emoji types and contextual conditions where the different types of emojis function more efficiently. We achieve these goals by performing the following analyses over large collections of Instagram posts: 1) investigating the linear or curvilinear relationship between emojis (texts) and consumer reactions and estimating the probability of consumer engagement, 2) segmenting emojis into emotional and informational emojis and observing their effects in conjunctions with texts, and 3) examining the meaningful conditions that content creators, social media users, and managers need to account for.

2.2. Hypothesis development

Emojis have been increasingly adopted by individuals and firms in an attempt to increase the engagement of their audiences. While including emojis in promotional communications has been known to have a positive influence on audiences' response (Das et al., 2019), little is known about the influence of emoji length. Unlike emojis, texts have been heavily studied in social media contexts. Among various text-related features, the influence of text length has been examined in various studies, and many of them reveal its negative impact (Alboqami et al., 2015; Baltas, 2003; De Vries et al., 2012), which may be caused by readers' information overload. We expect that the similar phenomenon occurs with emojis in brand-related UGC as they are part of information in the setting. Based on this, we suggest the following two hypotheses:

H1. Including emojis in brand-related UGC is positively associated with consumer engagement.

H2. A number of emojis in brand-related UGC is negatively associated with consumer engagement.

Emotion-related features in online contents are relatively heavily studied, and the findings are largely context-dependent (Chen et al., 2020; Lee et al., 2018; Tellis et al., 2019). For example, Tellis et al. (2019) reveal that emotion-related ads are shared more on general social media platforms such as Facebook or Google + than on a professional networking service such as LinkedIn whereas the influences are opposite for information-related ads. Lee et al. (2018) study Facebook ads and find the significant and positive effect of brand-related contents (e.g., emotion and humor) on consumer engagement. Unlike these studies, our focus is brand-related UGC, but we expect that the positive impact of emotional content including emotional emojis holds for the brand-related UGC as the goal of the brand-related UGC is usually either general or entertaining rather than functional. These arguments lead us to suggest that:

H3. Emotional emojis in brand-related UGC are positively associated with consumer engagement.

In subsequent, we examine contextual conditions on which the impact of emotional emojis is contingent. In social media posts, emojis are frequently coupled with texts, and the related natural research question could be if there is a synergy between emojis and texts. In particular, for emotional emojis, we expect a positive interaction effect between a number of emotional emojis and text valence, because emotional emojis are frequently used to express users' emotions (Boutet et al., 2021), and content creators usually convey their positive emotions rather than negative ones through their posts in a brand-related social media setting. Based on the arguments, we expect a synergy between emotional emojis and text valence. That is, emotional emojis that are incorporated in positive text messages than in negative text messages are more likely to increase consumer engagement in brand-related social media posts. This reasoning leads us to suggest the following hypothesis:

H3a. Emotional emojis in brand-related UGC are positively associated with consumer engagement more in positive texts than in negative texts.

Advertising research reveals that emotions are positively associated with consumer responses. Tellis et al. (2019), for example, empirically show a positive association between emotional ads and virality in major social media platforms. Lee et al. (2018) reveal that emotions and humor on social media advertisements have a positive impact on consumer engagement. Prior to these findings with observational data, the positive association between emotions and consumer responses in commercial contexts are shown in numerous studies with a controlled setting (Cho & Schwarz, 2006; Gorn et al., 1993; Schwarz & Clore, 1983). This phenomenon occurs probably because consumers usually do not expect much emotion or humor from commercial contents compared with general posts based on one of the important goals of advertisements, which is conveying product information. However, when consumers face emotion or humor in the commercial content, it may cause consumers to have extra positive feelings toward the content, because the actual pleasance caused by the emotion-related content in commercial posts surpasses the expected one. On the other hand, one of the biggest utilities from general social media posts is entertainment rather than an information conveyance so emotion-related contents may not be surprising and thus do not have a significant impact on the audience of the content. Considering these, we predict that:

H3b. Emotional emojis are positively associated with consumer engagement more in commercial posts than in general posts.

Informational contents are usually dry and uninteresting based on their focus on arguments or factual descriptions and thus associated with consumers' negative responses to social media posts (Tellis et al., 2019). Aligning with this finding, we argue that informational emojis are negatively associated with consumer engagement in brand-related social media posts as consumers may feel bored or irritated by this type of content. This reasoning suggests the following hypothesis:

H4. Informational emojis in brand-related UGC are negatively associated with consumer engagement.

Unlike emotional emojis, informational emojis may have a detrimental influence on consumer engagement when coupled with positive text messages in brand-related social media posts as there may be no synergy between informational emojis and text valence due to the dry and factual focus nature of informational emojis. We thus expect that:

H4a. Informational emojis in brand-related UGC are negatively associated with consumer engagement more in positive texts than in negative texts.

Although we generally predict that informational emojis are negatively associated with consumer engagement in brand-related UGC, this influence may be contingent on post types as expected in H3b. General social media posts frequently use emotion-related contents, because image-related utility associated with a sense of self-worth is large for most users (Toubia & Stephen, 2013) and a self-expression purpose is significant for them when they create social media posts without commercial goals. Emotion-related contents are thus generally expected to appear in general social media posts. However, the appearance of informational contents in general social media posts may give a surprising pleasance to readers. With this reasoning, we expect that informational contents are positively associated with consumer engagement more in general social media posts than in commercial posts. Considering these factors, we predict that:

H4b. Informational emojis in brand-related UGC are positively associated with consumer engagement more in general posts than in commercial posts.

Finally, we develop our conceptual framework in Fig. 2, indicating the expected signs of the hypotheses that are described above.

3. Data

3.1. Data overview

We use Instagram as our data source for the analyses. When scraping Instagram posts, we employ brand names as source tags. For example, hashtags with brand names, such as Compaq or Nike, are used as source tags. The post that is extracted with each source tag should contain at least one brand name as one of its hashtags (e.g., *#compaq*, *#nike*).

To select the source tags, we start with a brand name list published by Lovett et al. (2014) that includes almost 700 leading national brands. Wanting to only include product brands from the list, we filter the list based on the variable *Product*. Next, we filter out the words that correspond to movie names (e.g., *The Lion King*, *Wonder Woman*) and brands with common-noun names (e.g., Brother, Degree). We then randomly select the brands from the remaining brand list and arrive at 86 brands. The 86 brands are the focus of our study.

When obtaining the Instagram data, we first collect the links to the posts via the application programming interface (API) that Instagram uses to display post previews on its tag listing pages (e.g., *#chanel*; instagram. com/explore/tags/chanel/). Prior to conducting the analyses, we preprocess the raw data using different methods. First, we filter out the posts containing description languages other than English. Then, we manually remove the posts that are created by brands (i.e., brand accounts) to ensure that our data only contain UGC. In doing so, we manually check the URLs of the 86 brand accounts and delete the posts that are associated with the brand accounts. Table 1 presents a summary of the constructs and variables that are incorporated into the models and provides their definitions.

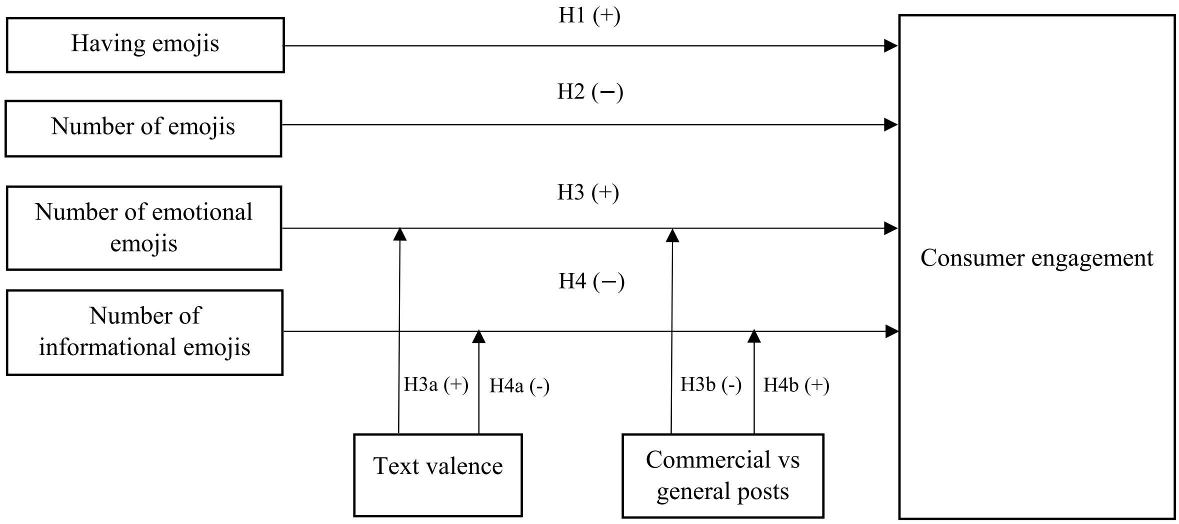


Fig. 2. Conceptual framework.

Table 1
Constructs, Variables, and Definitions.

Constructs	Variables	Definitions
Consumer Engagement		
Number of likes	*NumLikes*	Number of likes of a post
Number of comments	*NumComments*	Number of comments of a post
Emoji Variables		
Contain emojis	HasEmoji	Indicator of whether the post contains an emoji
Emoji length	EmojiLength	Number of emojis in a post
Number of emotional emojis	Emoji:Emo	Number of emotional emojis in a post
Number of informational emojis	Emoji:Info	Number of informational emojis in a post
Emoji popularity	EmojiPop:High EmojiPop: Medium EmojiPop:Low	Indicators of whether the post contains an emoji that belong to a high-usage group, a medium-usage group or/and a low-usage group
Text Variables		
Text valence	TextVal	Valence of texts including a mention and hashtags
Text divergence	TextDiv	Magnitude of emotions in texts including a mention and hashtags
Text length	TextLength	Number of texts including a mention and hashtags
Control Variables		
Post-Level Controls:		
General posts	GeneralPost	Indicator of whether a post is a general post or a commercial post
Number of faces	FaceCount	Number of faces in a post
Image quality	ImageQual	Perception based image quality
Object types	Living Food Plant	Indicators of whether the image contains a living thing; a food product; a plant, respectively
Age of a post	PostAge	Age of a post since being uploaded in days
Time of a day of a post	TimeDay	Integer indicating hour in a day (0 = 12:00 a.m., 1 = 1:00 a.m., …, 23 = 11:00 p.m.)
Contains punctuation	HasPunctuation	Indicator of whether a post contains either question marks or exclamation marks
User-Level Controls:		
Number of followers	FollowerCount	Number of followers of a user
Number of followings	FollowingCount	Number of followings of a user
Number of posts	PostCount	Number of posts of a user

3.2. Variable definitions and manipulations

We utilize various techniques or sources, such as computer vision API, natural language processing (NLP), regular expressions, or a Unicode list to extract and manipulate the emoji and text variables and the control variables, including image characteristics, time characteristics, and user characteristics.

3.2.1. Emoji variables

Emojis (HasEmoji): We label posts based on whether a post contains emojis and create a binary variable.

Number of emojis (EmojiLength): Using the emoji list obtained from the Unicode® Consortium1, the governing body for maintaining and publishing digital characters, regular expression is created to detect each emoji's unique code-points. Using the regular expression, the emojis used in each Instagram post are identified and placed into a list where its length could be counted. The value of the length of each list gives the number of emojis used per post.

Number of emotional, informational emojis (Emoji:Emo, Emoji:Info): Each emoji is classified into emotional and informational categories based on the names of pre-existing groups set by the Unicode® Consortium. Then, each emoji, placed in the two, newly created groups, is individually reviewed to ensure that its meaning is consistent with the group it is in.[2] To illustrate, all the emojis in the pre-existing group, 'Flags', are assumed to convey information to the reader. Emojis in the 'Smilies and Emotion' group are assumed to be emotional in nature. After the initial categorization process, the emojis in the informational and emotional groups are individually analyzed to ensure that any outlier emojis (i.e., ambiguous emojis that do not seem to belong in either group) are removed from the group. For instance, the pre-existing group, 'People & Body', consists of emojis that are intrinsically informational. However, while some emojis can be categorized into an emotional group depending on the context in which they are used, they are still grouped under 'People & Body'. It is important to note that the number of emojis is skewed toward the informational group in comparison to the number of emojis in the emotional group. Examples of emotional and informational emojis are displayed in Fig. 3.

Emoji popularity (EmojiPop:High, EmojiPop:Medium, EmojiPop:

[2] The pre-existing group that is categorized into an emotional emoji cluster is 'Smileys & Emotion', and the groups that are categorized into an informational emoji cluster are 'People & Body', 'Animals & Nature', 'Food & Drink', 'Travel & Places', 'Activities', 'Objects', 'Symbols', and 'Flags'.

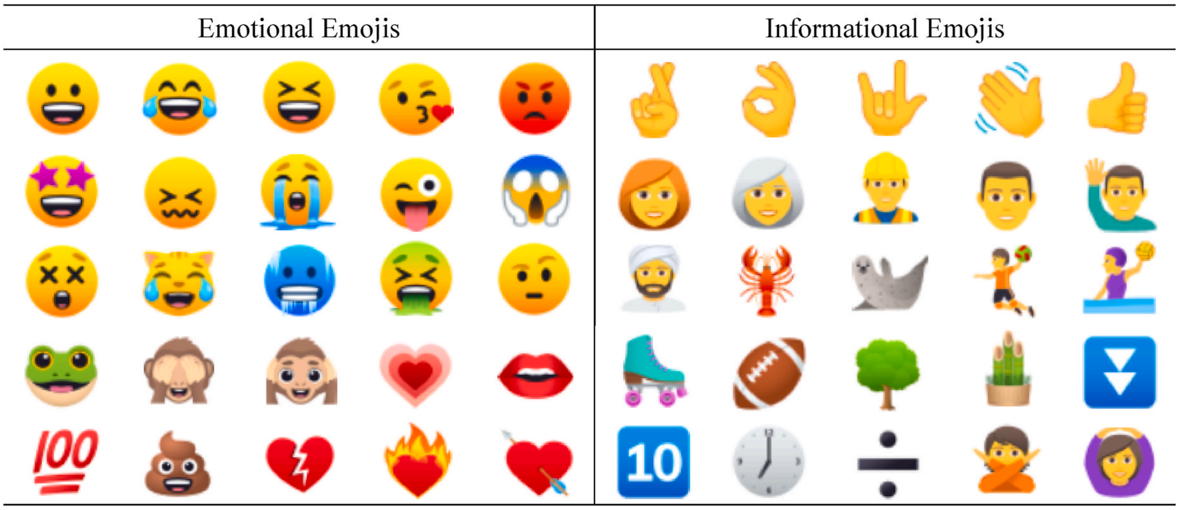


Fig. 3. *Emotional vs. Informational Emojis.*

Low): A total of 1456 unique emojis are found in the data set. To begin, we divide each emojis into three groups based on the frequency of their usage (e.g., high usage, intermediate usage, low usage). The first 486 most frequently used emojis are placed in the high-usage group (EmojiPop:High). The next 485 emojis are grouped in the intermediate-usage group (EmojiPop:Medium) and the remaining 485 emojis are classified under the low-usage group (EmojiPop:Low). Then, a regular expression is created to detect which frequency group each emoji used in an Instagram post belongs to. Fig. 4 exhibits the examples that fall into each category based on popularity.

3.2.2. Text variables

Text valence (TextVal) and Text divergence (TextDiv): We use SentiStrength (Thelwall et al., 2010) to capture the polarity aspects of text sentiment (TextVal) and text divergence (TextDiv). We compute text valence by adding the two values ($= positive + negative$) and text divergence using the formula $sentiment = (positive - negative) - 2$, following Stieglitz and Dang-Xuan (2013).

Number of texts (TextLength): A regular expression is used to detect the number of texts based on white space. For example, a string that reads “HelloGoodbye” is counted as one word by the regular expression, whereas “Hello Goodbye” is detected as two words.

3.2.3. Post-level controls

General posts (GeneralPost): Our data do not contain advertisements generated by firms or organizations as we delete all the posts associated with firms’ brand accounts. However, even without the advertisements published by firms or organizations, our data may still contain commercial posts generated by small businesses or individual sellers as social media platforms are often used as a medium of direct selling or individual trading. These commercial posts should be distinguished from the general posts without commercial purposes, because the content in these commercial posts might be characterized differently from the one of general posts. For example, this type of commercial posts is focused more on information conveyance for direct selling so displays simple images focusing on product details without too much background objects. Text contents are also focused on highlighting product features (e.g., price, size, color), again, because the goal of the commercial posts is direct selling or individual trading, not building brand preference or brand loyalty. Due to the distinguished nature of the commercial posts in the content and the goal, emojis may function differently across the two different types of UGCs (commercial posts vs. general posts).

We use a user-based method to distinguish general posts from commercial posts. To detect users’ commercial purposes, we use the Instagram user profiles (user descriptions) employing the following steps. First, we randomly select 1000 user IDs and the user profiles associated with the IDs. Then, we create a dictionary of words that reveals the users’ retailing-related purposes in their user profiles. For example, some user profiles disclose a seller’s contact information (e.g., WhatsApp), and other profiles directly reveal their retailing purposes using words such as “discount”, “preorder”, “shopper”, “service”, or “shipping”. Through this manual process, we find 209 commercial-related words from the 1000 user descriptions. We, then, detect all the users whose user profiles include at least one commercial-related word and label the posts associated with these users as commercial posts. Finally, we manually check the accuracy of the labels (a commercial post vs. a general post) with 1000 randomly selected posts and find that 86% of the posts are correctly categorized. The commercial post explains 24% of

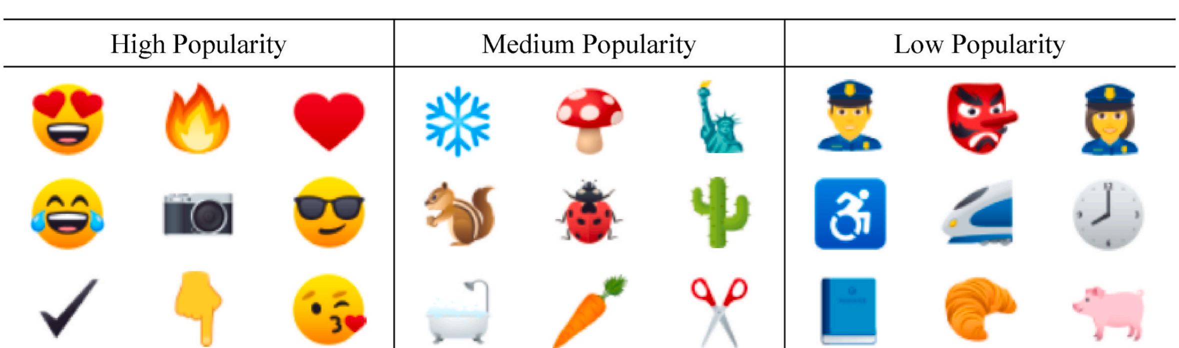


Fig. 4. *Examples of Emojis based on Popularity.*

our data.

Number of faces (FaceCount): Prior literature shows that faces engage people in online social media (Bakhshi et al. 2014) and are thus controlled in our model. The number of faces is an integer value that counts how many faces exist in each image and is directly extracted with the computer vision API.

Image quality (ImageQual): Image properties are known to impact on consumer responses (Zhang et al., 2021), and image quality is incorporated as a control variable in our model. The image quality is measured with Perception-based Image Quality Evaluator (PIQE). It is an unsupervised algorithm and returns a nonnegative scalar in the range [0,100]. The returned measure is inversely correlated with the perceptual quality of an image, so a low value indicates high perceptual quality and a high value indicates low perceptual quality.

Object types (HasLiving/HasFood/HasPlant): Computer vision API extracts object names in an image as a noun form. The object names are our main data source to create the labels for the object types. We start by creating a list of unique nouns from the object names. We then apply *WordNet*, a large lexical database of English based on a hierarchical structure, to the unique noun list to compute the semantic similarities between the nouns in the list and a distance matrix. Hierarchical clustering with Ward's method is then employed to cluster the nouns into similar semantic groups. Through this process, we find five groups (*living*, *non-living*, *food and plants*, *scenery and events*, and *adjectives*) that are semantically similar. We discard the *adjectives* group, because *adjectives* usually do not indicate objects or things. Moreover, we do not use *non-living* and *scenery and events* in the further analyses as there are few variations in the variables. Finally, we divide *food and plants* into two groups (*food* and *plants*), and use them for our analyses.

Age of a post (PostAge): The age of a post indicates the number of days since a post was published on Instagram. This variable is computed from the time stamps, which is one of the variables that is directly retrieved from the Instagram API.

Time of a post (TimeDay): Time of a day when a post was uploaded is coded as hours passed since midnight (0–23).

Punctuation (HasPunctuation): We detect whether a post contains either question marks or exclamation marks using regular expression, and we code the variable in a binary manner.

3.2.4. User-level controls

Numbers of followers/followings/posts (FollowerCount/FollowingCount/PostCount): We control for the number of followers (number of users who follow a post creator), the number of followings (number of users who are followed by a post creator), and the number of posts (number of posts that a post creator publishes). These variables are directly extracted through Instagram API.

3.3. Descriptive statistics

In Table 2, we report the descriptive statistics of both the continuous and categorical variables and summarize the continuous variables in terms of mean, median, standard deviation, minimum, maximum, and skewness and the categorical variables in terms of frequency, the number of individual observations in any given category. In addition to the descriptive statistics, we look at the correlations among the continuous variables. While the analysis reveals low correlations for most of the relationships, high correlations are found between the number of likes and the number of comments (0.64) and between the number of informational emojis and emoji length (0.93).

4. Empirical analysis and results

We estimate the characteristics of consumer engagement (*NumLikes, NumComments*) using the following fixed-effect model. Endogeneity may be a concern when assessing the relationship between independent variables in the models and consumer engagement. In particular, we are concerned about the selection bias coming from brand names, so we employ a brand fixed-effect model to alleviate this issue.

We estimate the following equation for $NumLikes_{ij}$ as a dependent

Table 2
Descriptive Statistics.

Variable(s)	Continuous							Categorical	
	N	Mean	Std Dev	Min	Median	Max	Skewness	Yes	No
Consumer Engagement									
NumLikes	31,621	169.16	920.49	0.00	32.00	34,494.00	15.62		
NumComments	31,621	3.21	9.41	0.00	1.00	120.00	7.45		
Emoji or Text Variables									
HasEmoji	31,621	0.29	0.46	0.00	0.00	1.00	0.90	9293	22,328
EmojiLength	31,621	1.04	3.04	0.00	0.00	108.00	8.40		
Emoji:Emo	31,621	0.30	1.08	0.00	0.00	42.00	12.00		
Emoji:Info	31,621	0.70	2.48	0.00	0.00	101.00	10.36		
EmojiPop:High	31,621	0.28	0.45	0.00	0.00	1.00	0.97	8897	22,724
EmojiPop:Medium	31,621	0.07	0.25	0.00	0.00	1.00	3.44	2147	29,474
EmojiPop:Low	31,621	0.03	0.16	0.00	0.00	1.00	5.96	821	30,800
Text Variables									
TextVal	31,621	0.68	1.54	−7.00	0.00	8.00	0.10		
TextDiv	31,621	4.07	1.75	2.00	4.00	14.00	0.80		
TextLength	31,621	23.41	33.30	1.00	12.00	409.00	3.31		
Control Variables									
Post-Level Controls:									
GeneralPost	31,621	0.76	0.43	0.00	1.00	1.00	−1.19	23,878	7743
FaceCount	31,621	0.38	0.99	0.00	0.00	21.00	5.49		
ImageQual	31,621	42.67	12.67	5.60	42.38	88.85	0.15		
Living	31,621	0.56	0.50	0.00	1.00	1.00	−0.24	17,705	13,916
Food	31,621	0.32	0.47	0.00	0.00	1.00	0.78	10,034	21,587
Plant	31,621	0.14	0.35	0.00	0.00	1.00	2.09	4382	27,239
PostAge (days)	31,621	43.29	149.18	2.00	7.00	2421.00	8.11		
TimeDay (hours)	31,621	11.92	7.63	0.00	13.00	23.00	−0.11		
HasPunctuation	31,621	0.22	0.41	0.00	0.00	1.00	1.35	6955	24,666
User-Level Controls:									
FollowerCount	31,621	6627.29	58,488.82	0.00	571.00	4,462,685.00	42.84		
FollowingCount	31,621	980.79	1536.85	0.00	416.00	8162.00	2.88		
PostCount	31,621	1049.76	3008.83	0.00	337.00	87,697.00	10.55		

variable:

$$\ln\left(NumLikes_{ij}\right) = \alpha_i + x_{ij}^{'}\beta + \varepsilon_{ij}, \quad (1)$$

in which $NumLikes_{ij}$ indicates (log of) number of likes for post i of user j; α_i is a brand fixed effect; and x_{ij} is vector of post and user characteristics including emoji variables (HasEmoji, EmojiLength, Emoji:Emo, Emoji:Info, EmojiPop:High, EmojiPop:Medium, EmojiPop:Low), text variables (TextLength, TextVal, TextDiv), post-level controls (FaceCount, ImageQual, HasLiving, HasFood.

HasPlant, PostAge, TimeDay, HasPunctuation), user-level controls (FollowerCount, FollowingCount, PostCount), and the interaction variables. The same model is used for (log of) number of comments.

4.1. Influence of emojis vs. texts on consumer engagement

We study the main effects of the emoji and text characteristics on consumer engagement, and Table 3 presents the estimated coefficients for *NumLikes* and *NumComments,* respectively (*LIKES* Model & *COMMENTS* Model). In the models, log of (*NumLikes*) and log of (*NumComments*) are regressed on the emoji variables, the text variables, and the control variables, including the post-level controls and the user-level controls. The third and fifth columns in Table 3 present the effect of each independent variable in terms of percentages.

We note several results. First, having emojis (HasEmoji) is positively associated with consumer engagement (*NumLikes*: 0.54, $p < .000$; *NumComments*: 0.53, $p < .000$) while greater use of emojis (EmojiLength) is negatively associated with consumer engagement (*NumLikes*: −0.01, $p < .000$; *NumComments*: −0.02, $p < .000$), which is consistent with H1 and H2. This implies that including emojis in brand-related UGC is likely to improve consumer engagement, but having too many emojis can have a negative influence on consumer engagement. In terms of percentage change, including emojis is associated with 72% increase in the average number of likes and 70% increase in the average number of comments in comparison to posts without any emojis. Although they are not our focal variables, the analysis reveals several interesting findings from our control variables. There is a consistent and non-linear (concave) relationship between text length and consumer engagement, such that the parameter of text length is positive and significant (*NumLikes*: 0.01, $p < .000$; *NumComments*: 0.02, $p < .000$), and the parameter of text length squared (TextLengthˆ2) is negative and significant (*NumLikes*: −0.00005, $p < .000$; *NumComments*: −0.00007, $p < .000$). This implies that consumer engagement increases with text length, but the relationship becomes negative when a post contains too many words and hashtags. The economic interpretation of this result would be: for one unit increase in the average number of texts, the average number of likes (comments) increases by 1% (2%) up to the maximum level of consumer engagement (100 text length for *NumLikes* and 145 text length for *NumComments),* and then the average number of likes (comments) decreases by 0.005% (0.01%). Finally, HasPunctuation is positively related with consumer engagement. This implies that having either a question mark or an exclamation sign is related with more consumer engagement in brand-related UGC. The findings show that the average number of likes and the average number of comments increase by 16% and 28%, respectively, for a one unit increase in the average number of punctuations.

4.2. Emotional vs. informational emojis and their interactions with texts

In this section, we focus on two types of emojis (emotional emojis and informational emojis) and their interactions with text properties. We exclude HasEmoji and EmojiLength that are used for the models discussed in Section 4.1 due to the high correlation between EmojiLength and Emoji:Info (0.93). For the estimation, we, again, employ a brand fixed-effect model and keep all of the variables that are discussed in Section 4.1, except for the two emoji variables (HasEmoji and

Table 3
Estimated Effects of Emojis and Texts on NumLikes and NumComments.

	LIKES Model (1)	Percent Change (%) (2)	*COMMENTS* Model (3)	Percent Change (%) (4)
Emoji Variables				
HasEmoji	0.54***(.06)	71.60	0.53***(.04)	69.89
EmojiLength	−0.01***(.00)	−1.00	−0.02***(.00)	−1.98
EmojiPopHigh	0.42***(.02)	52.20	0.39***(.02)	47.70
EmojiPopMedium	0.04(.03)	4.08	0.02(.03)	2.02
EmojiPopLow	−0.13**(.04)	−12.19	−0.15**(.04)	−13.93
Text Variables				
TextLength	0.01***(.00)	1.01	0.02***(.00)	2.02
TextLengthˆ2	−0.00005***(.00)	−0.005	−0.00007***(.00)	−0.01
TextVal	0.01***(.00)	1.01	−0.004(.00)	−0.40
TextDiv	0.01**(.00)	1.01	−0.04***(.00)	−3.92
Control Variables				
Post-Level Controls:				
FaceCount	0.06***(.01)	6.18	0.03***(.00)	3.05
ImageQual	−0.001*(.00)	−0.10	−0.001*(.00)	−0.10
HasLiving	0.20***(.01)	22.14	0.08***(.01)	8.33
HasFood	−0.06***(.01)	−5.82	−0.02#(.01)	−1.98
HasPlant	0.15***(.02)	16.18	0.02#(.01)	2.02
PostAge (days)	−0.0004***(.00)	−0.04	0.0001**(.00)	0.01
TimeDay (hour)	0.006***(.00)	0.60	0.002***(.00)	0.20
HasPunctuation	0.15***(.02)	16.18	0.25***(.01)	28.40
User-Level Controls:				
FollowerCount	0.000006***(.00)	0.00	0.000003***(.00)	0.00
FollowingCount	0.00004***(.00)	0.00	0.000003(.00)	0.00
PostCount	−0.000009***(.00)	0.00	−0.00001***(.00)	0.00
Brand Fixed-Effects	Yes		Yes	
Pseudo R-Squared	.32		.32	
AIC	96,758.61		96,758.61	
N	31,621		31,621	

#$p < .05$; *$p < .01$; **$p < .001$; ***$p < .0001$.

EmojiLength). Using the set of variables, we estimate the main effects and interaction effects between the two types of emoji variables and text variables for *NumLikes* and *NumComments*. In Table 4, we report the estimated results.

Main effects. Columns (1) and (3) display the estimation results both with main effects and with the interaction effects for *NumLikes* and *NumComments*, respectively. We note that the main findings remain consistent with the results reported in Section 4.1; the text length shows a curvilinear relationship with consumer engagement and having either question marks or exclamation marks is associated with more consumer engagement. For the two types of emoji variables that are newly added in this model, we find that the length of emotional emojis (Emoji:Emo) does not reveal a clear association with consumer engagement, which rejects H3. The length of informational emojis (Emoji:Info) is, however, strongly and negatively related with consumer engagement (*NumLikes*: -0.02, $p < .000$; *NumComments*: -0.03, $p < .000$), confirming H4. This suggests that excessive use of informational emojis may hurt consumer engagement in brand-related UGC. Economically, the result shows that the average number of likes and the average number of comments decrease by 2% and 3%, respectively, for a one unit increase in the average number of informational emojis in brand-related UGC.

*Emoji types*Text valence*. More interestingly, there is a significant and positive interaction effect between the number of emotional emojis and text valence (columns (1) and (3) in Table 4). There appears to be a synergy between emotional emojis and text valence (*NumLikes*: 0.01, $p < .000$; *NumComments*: 0.006, $p < .05$), implying that the more positive the text in brand-related UGC, the stronger the positive influence of emotional emojis on consumer engagement. This suggests that having more emotional emojis can help a post receive more likes or comments when the text sentiment is positive (H3a). However, having informational emojis can be detrimental to consumer engagement in a similar social media post if the post contains text with a positive sentiment (*NumLikes*: -0.004, $p < .10$; *NumComments*: -0.003, $p < .01$) (H4a).

*Emoji types*Text divergence.* In addition to the interaction term between emotional emojis and text valence, we observe the interaction effect between emoji types and text divergence though they are not our focal variables. Columns (1) and (3) in Table 4 exhibit the estimated results between emoji types and text divergence. There is a negative and significant interaction effect between the length of the emotional emojis and text divergence (*NumLikes*: -0.01, $p < .01$; *NumComments*: -0.005, $p < .05$). This implies that having an excessive number of emotional emojis in the social media posts containing texts being skewed toward extreme emotions may be detrimental for a social media post, impeding its ability to receive more likes or comments.

In Fig. 5, we visually highlight the interaction effects between the two types of emojis and text valence, which we consider to be noteworthy. The plot (A) and plot (B) in Fig. 5 display the interaction effects between the length of emotional emojis and text valence for *NumLikes* and *NumComments*, respectively, where the three points in the legend represent one standard deviation above the mean (+1SD), mean, and one standard deviation below the mean (-1SD). For the emotional emojis (Emoji:Emo × TextVal), we find a positive trend between text valence and fitted values for +1SD (=1.38) and a slightly negative trend between them for -1SD (=-0.78). The trends are opposite for the informational emojis (Emoji:Info × TextVal). Plot (D) in Fig. 5 displays a negative trend between text valence and fitted values for +1SD (=3.18) and a positive trend between them for -1SD (=-1.78).

Table 4
Estimated Interaction Effects between Emojis and Texts.

	LIKES Model	Percent Change (%)	*COMMENTS* Model	Percent Change (%)
	(1)	(2)	(3)	(4)
Emoji Variables				
EmojiEmo	0.02(.02)	2.02	0.02#(.01)	2.02
EmojiInfo	−0.02***(.01)	−1.98	−0.03***(.00)	−2.96
EmojiPopHigh	0.50***(.02)	64.87	0.46***(.01)	58.41
EmojiPopMedium	0.07**(.03)	7.25	0.07***(.02)	7.25
EmojiPopLow	−0.13**(.04)	−12.19	−0.14***(.03)	−13.06
Text Variables				
TextLength	0.01***(.00)	1.01	0.02***(.00)	2.02
TextLength^2	−0.00005***(.00)	−0.005	−0.00007***(.00)	−0.01
TextVal	0.01**(.00)	1.01	−0.005(.00)	−0.50
TextDiv	0.01**(.00)	1.01	−0.04***(.00)	−3.92
Interaction Effects				
EmojiEmo:TextVal	0.01***(.00)	1.01	0.006*(.00)	0.60
EmojiEmo:TextDiv	−0.01**(.00)	−1.00	−0.005*(.00)	−0.50
EmojiInfo:TextVal	−0.004#(.00)	−0.40	−0.003**(.00)	−0.30
EmojiInfo:TextDiv	0.001(.00)	0.10	0.002*(.00)	0.20
Control Variables				
Post-Level Controls:				
FaceCount	0.06***(.01)	6.18	0.03***(.00)	3.05
ImageQual	−0.001*(.00)	−0.10	−0.0007*(.00)	−0.07
HasLiving	0.20***(.01)	22.14	0.08***(.01)	8.33
HasFood	−0.06***(.01)	−5.82	−0.02#(.01)	−1.98
HasPlant	0.15***(.02)	16.18	0.02#(.01)	2.02
PostAge (days)	−0.0004***(.00)	−0.04	0.0001**(.00)	0.01
TimeDay (hour)	0.006***(.00)	0.60	0.002***(.00)	0.20
HasPunctuation	0.15***(.02)	16.18	0.25***(.01)	28.40
User-Level Controls:				
FollowerCount	0.000006***(.00)	0.00	0.000003***(.00)	0.00
FollowingCount	0.00004***(.00)	0.00	−0.00008**(.00)	0.00
PostCount	−0.000009***(.00)	0.00	−0.00001***(.00)	0.00
Brand Fixed-Effects	Yes		Yes	
Pseudo R-Squared	.32		.41	
AIC	96,875.44		66,856.61	
N	31,621		31,621	

#$p < .05$; *$p < .01$; **$p < .001$; ***$p < .0001$.

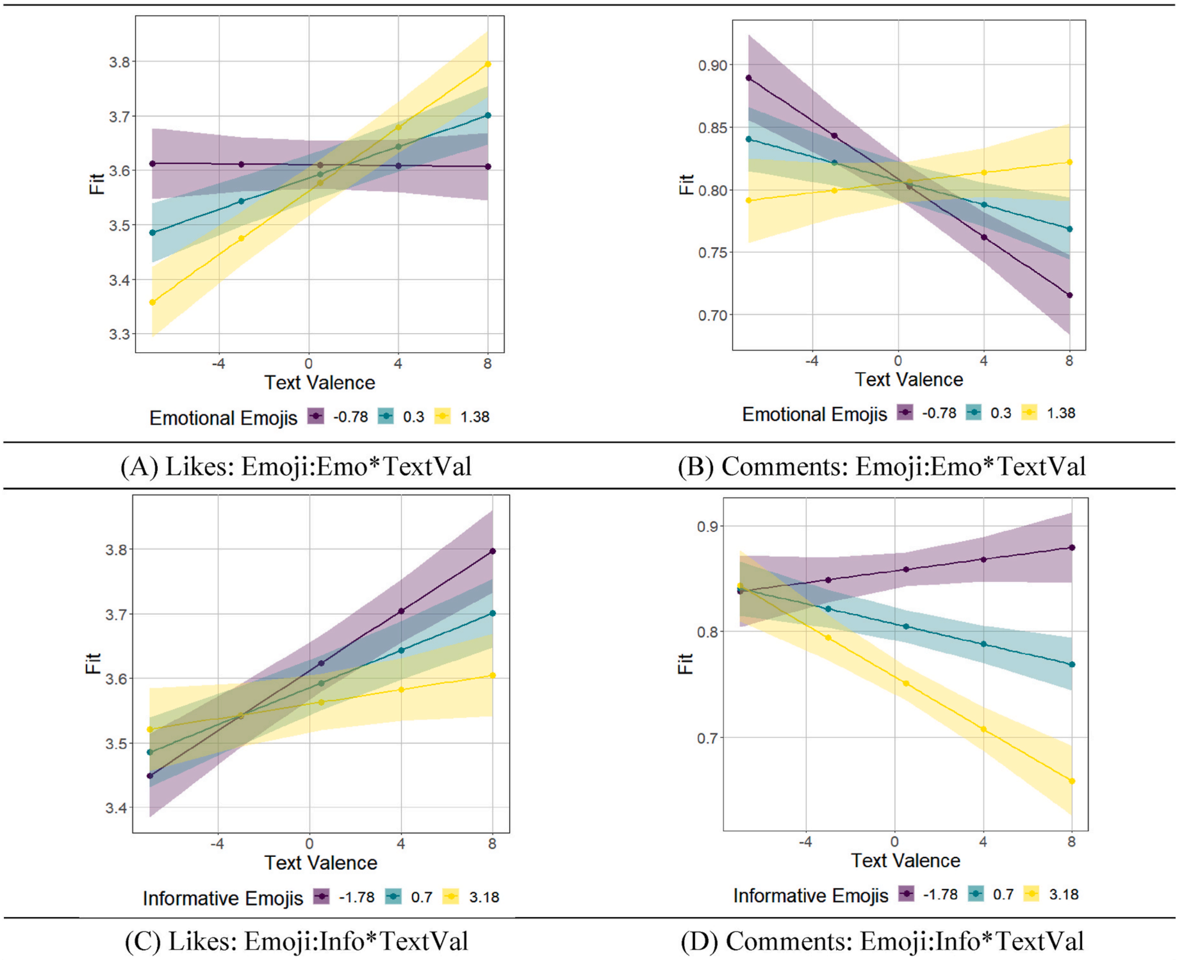


Fig. 5. *Interaction Effects: Emojis and Texts.*

4.3. Contextual conditions affecting the influence of emojis on consumer engagement

In this section, we aim to replicate the findings presented in Section 4.1 and Section 4.2 and to investigate the contextual conditions that affect the influence of emojis on consumer engagement in social media posts. We include one focal contextual condition (GeneralPost) and two more contextual variables (HasPunctuation; TimeDay) that may mediate the emoji impact on consumer engagement. We estimate the two models using a brand fixed-effect model. Table 5 presents the result. We estimate two separate models (Model 1 and Model 2) where we incorporate HasEmoji and EmojiLength in Model 1 and Emoji:Emo and Emoji:Info in Model 2. We estimate the two models separately due to the high correlation between EmojiLength and Emoji:Info. We observe that the main effects that are presented in Section 4.1 are replicated (Model 1 of Table 5). Again, we find that there is a significant main effect for the presence of an emoji (*NumLikes*: 0.54, $p < .000$), but having too many emojis is negatively related with consumer engagement (*NumLikes*: $-0.01, p < .000$). In Model 2, the result again reveals that the number of informational emojis is negatively associated with consumer engagement in brand-related UGC (*NumLikes*: -0.02, $p < .01$). The inverted U relationship between text length and consumer engagement is replicated in both Model 1 and Model 2. Below, we describe the estimation results that are noteworthy from these two models.

*Length of emotional emojis *General post.* Having more emotional emojis is associated with a lower level of consumer engagement when brand-related UGC is created by general users; however, it is associated with an increase in consumer engagement in commercial posts, which is consistent with H3b. The coefficient of the length of emotional emojis (0.02, not significant at $p < .10$) is the slope of the regression line for commercial posts (reference group). The value for the general posts by the length of emotional emoji interaction is -0.05 ($p < .01$), and this results in -0.03 (=0.02–0.05) for the slope of general posts. The positive trend for commercial posts and the negative trend for general posts are displayed in Plot (A) in Fig. 6. The plot shows that, for commercial posts, consumer engagement shows up-trend as a social media post contains more emotional emojis. However, for general posts, consumer engagement displays a down-trend if a social media post contains too many emotional emojis. This result suggests that having more emotional emojis may be beneficial for the social media users with commercial purposes, while general social media users need to be more cautious in overly using emotional emojis in their posts. This is surprising because: 1) advertisements with commercial purposes generally focus more on information rather than emotion-related features and 2) many general social media users have self-expression purposes, and they frequently use emotion-related features to express themselves. In terms of percentage change, for a one unit increase in the average number of emotional emojis, the average number of likes decreases by 3% for general posts and increases by 2% for commercial posts.

*Length of informational emojis*General post.* The coefficient of the length of informational emojis (-0.02, $p < .01$) is the slope of the regression line for commercial posts (reference group). Plot (D) in Fig. 6 displays the interaction effect. The value for the general posts by Emoji:Info interaction is 0.01 ($p < .05$). The slope for the general posts is -0.01 (=-0.02 + 0.01). The negative effect of informational emojis is greater for commercial posts than for general posts, which supports H4b.

Table 5
Estimated Effects of Contextual Conditions.

	LIKES	Percent	*LIKES*	Percent
	Model1	Change (%)	Model2	Change (%)
	(1)	(2)	(3)	(4)
Emoji Variables				
HasEmoji	0.54***(.07)	71.60		
EmojiLength	−0.01***(.00)	−1.00		
Emoji:Emo			0.02(.02)	2.02
Emoji:Info			−0.02**(.01)	−1.98
EmojiPop:High	0.02(.06)	2.02	0.49***(.02)	63.23
EmojiPop:Medium	−0.02(.03)	−1.98	0.06*(.03)	6.18
EmojiPop:Low	−0.18***(.04)	−16.47	−0.13**(.04)	−12.19
Text Variables				
TextLength	0.01***(.00)	1.01	0.01***(.00)	1.01
TextLength^2	−0.00004***(.00)	−0.004	−0.00005***(.00)	−0.005
TextVal	0.02***(.00)	2.02	0.02***(.00)	2.02
TextDiv	0.01***(.00)	1.01	0.01**(.00)	1.01
Interaction Variables				
HasEmoji*GeneralPost	0.02(.03)	2.02		
HasEmoji*HasPunctuation	−0.30***(.03)	−25.92		
HasEmoji*TimeDay	0.003#(.00)	0.30		
Emoji:Emo*GeneralPost			−0.05**(.02)	−4.88
Emoji:Info*GeneralPost			0.01*(.01)	1.01
Emoji:Emo*HasPunctuation			−0.003(.01)	−0.30
Emoji:Info*HasPunctuation			−0.01*(.01)	−1.00
Emoji:Emo*TimeDay			0.0004(.00)	0.04
Emoji:Info*TimeDay			0.0003(.00)	0.03
Control Variables				
Post-Level Controls:				
GeneralPost	0.27***(.02)	31.00	0.28***(.02)	32.31
FaceCount	0.05***(.01)	5.13	0.06***(.01)	6.18
ImageQual	−0.001*(.00)	−0.10	−0.001*(.00)	−0.10
HasLiving	0.18***(.01)	19.72	0.18***(.01)	19.72
HasFood	−0.06***(.01)	−5.82	−0.06***(.01)	−5.82
HasPlant	0.14***(.02)	15.03	0.14***(.02)	15.03
PostAge (days)	−0.0004***(.00)	−0.04	−0.0004***(.00)	−0.04
TimeDay (hour)	0.005***(.00)	0.50	0.006***(.00)	0.60
HasPunctuation	0.30***(.02)	34.99	0.17***(.02)	18.53
User-Level Controls:				
FollowerCount	0.000006***(.00)	0.00	0.000006***(.00)	0.00
FollowingCount	0.00005***(.00)	0.01	0.00004***(.00)	0.00
PostCount	−0.000007#(.00)	0.00	−0.000008***(.00)	0,00
Brand Fixed-Effects	Yes		Yes	
Pseudo R-Squared	.32		.32	
AIC	96,654.86		96,569.79	
N	31,621		31,621	

#$p < .05$; *$p < .01$; **$p < .001$; ***$p < .0001$.

Economically, these results suggest that the average number of likes drops by 1% for general posts and drops by 2% for commercial posts for a one unit increase in the average number of informational emojis. This implies that social media posts with commercial purposes should be more cautious in using informational emojis than the social media posts without any business purposes.

*Emoji presence*Punctuation presence*. Beyond the focal contextual conditions that we are interested in, we further find several interesting contextual conditions that impact the relationship between emoji characteristics and consumer engagement. First, we find an interesting and significant interaction effect of the presence of emojis and the presence of punctuations. The result of Model 1 shows that the interaction effect is negative and statistically significant (*NumLikes*: −0.30, $p < .000$). To better understand what the interaction effect reveals, we create a plot that displays the fitted values of the dependent variable on the y-axis while the x-axis exhibits the values of the presence of punctuation (Plot (C) in Fig. 6). The two lines represent the values of the second independent variable, the presence of emojis. Plot (C) reveals that the condition where emojis are present is associated with a higher level of consumer engagement than the condition without emojis, whether or not punctuation is used. However, in the condition where there are no emojis, punctuation plays an important role; having punctuation in brand-related UGC is likely to increase consumer engagement.

*Emoji presence*Time of a day*. There is a marginal significance between time of a day and having emojis in brand-related UGC; the coefficient for TimeDay (*NumLikes*: 0.005, $p < .000$) is the slope of the regression line for the posts without emojis (reference group). The value for the posts with emojis by time of a day interaction is 0.003 ($p < .05$), which is the difference between the posts with emojis ("emoji group") and the posts without emojis ("no emoji group"). The slope for the "emoji group" is 0.008 (=0.005 + 0.003). The difference between "emoji group" and "no emoji group" is marginally and significantly different for different values of time of a day. This implies that when emojis exist in a post, consumer engagement more quickly grows as the time of a day increases from 0 (12:00 a.m.) to 23 (11:00 p.m.). However, the increasing trend is slower when emojis are not present in a social media post. If social media users would like to increase consumer engagement, both including emojis in their posts and publishing the posts late at night might be a beneficial strategy. In terms of percentage change, for a one unit increase in the time of a day, the average number of likes increases by 0.30% in brand-related social media posts without any emojis; it increases by 0.80% in social media posts containing emojis (Plot (D) in Fig. 6).

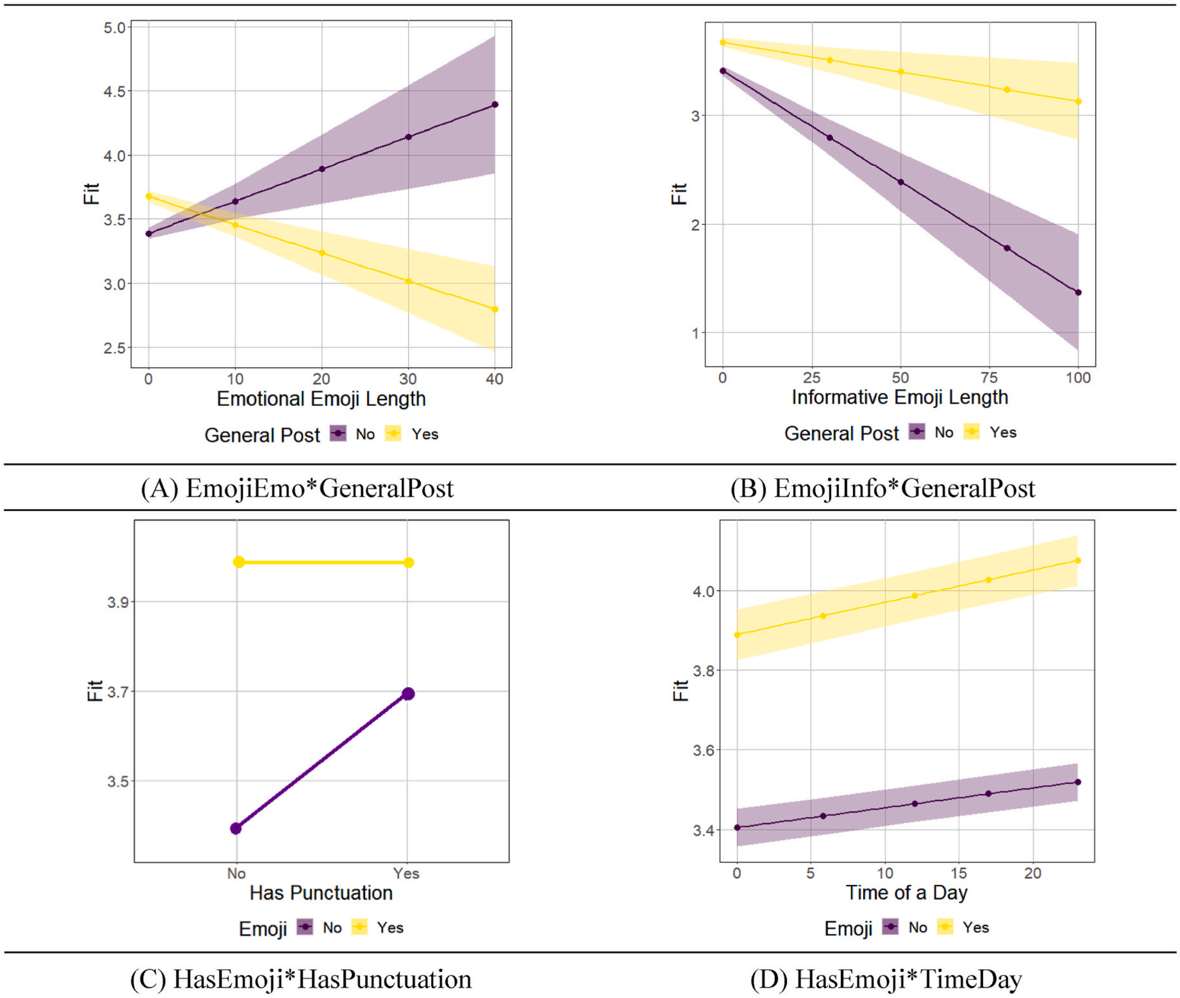


Fig. 6. *Interaction Effects: Emojis and Contextual Conditions.*

4.4. Robustness checks

4.4.1. Excluding extreme observations including key-opinion leaders (KOLs) and text sentiment variables

Although we hire a brand fixed-effect model to address selection bias, there might be other potential issues. For example, the influence of key opinion leaders (KOLs) may be much more significant than the one of others. In order to ensure the robustness of the models without such extreme observations, we trim 1% and 5% of each side of the data based on a number of follows. The findings largely remain the same to the ones in Sections 4.1, 4.2, and 4.3, revealing consistent results in terms of signs and magnitudes of the coefficients (Tables A1 & A2). We also conduct another robustness check without text valence and text divergence variables as they are moderately correlated (0.28). The results (Tables A3 & A4) remain unchanged when we exclude the two text variables.

4.4.2. Out-of-sample predictive model

We perform out-of-sample validations to test the assumptions of our models and forecast the performance. A logistic regression model is employed to test the predictive power of the factors that are incorporated into the models in Section 4.1, Section 4.2, and Section 4.3. The nature of the outcome variables (*NumLikes, NumComments)* is numeric, and we convert the numeric variables into binary variables by dividing the outcome variables into high and low consumer engagement based on the median of the distribution of likes and comments (Tellis et al., 2019). For the predictors, we use the same set of the variables that are employed in the models of Sections 4.1, 4.2, and 4.3. In Section 4.3, we estimate two different models (Model 1 and Model 2 in Table 5) using different sets of variables, so we perform the out-of-sample validations for the two different models. We use 80% of the data as a calibration sample and 20% as a holdout sample for all the validations. Accuracy, recall, and F1 score are used to evaluate the predictive performance; these are the most commonly used metrics to observe the forecasting power of validation models. Overall, the results indicate that the predictors of consumer engagement have high predictive power. The accuracy for *NumLikes* is about 72% for all the validation models (between 72% and 78% for recall and F1 scores), and the accuracy for *NumComments* ranges between 74% and 78.5% (between 69% and 82% for recall and F1). The high accuracy in the out-of-sample validation implies that the predictors or consumer engagement are potentially generalizable and not idiosyncratic. We further check the robustness of the predictive models under several different conditions to leverage the predictive value of the emojis and text variables and find that the accuracy drops from 5% to 10% without the emoji and the text variables or/and the interaction terms.

5. Discussion and conclusion

Using large-scale observational social media data, we examine how emojis are efficiently employed in brand-related UGC along with texts to make social conversations more engaging and the contextual conditions that affect the relationship between emojis and consumer engagement. We consider that our key findings are as follows. First, our main effects reveal a detrimental influence of excessive use of emojis on consumer engagement as well as the positive influence of their mere presence in social media. Second, more importantly, the analyses reveal essential contextual conditions that help social media users determine how to incorporate emojis more effectively. We group emojis into emotional emojis and informational emojis, and examine the main effects of the two types of emojis on consumer engagement as well as how they interact with two different text properties: text sentiment and text divergence. The results reveal that more emotional emojis are positively related with consumer engagement only when the sentiment of the text portion of a social media post is positive. However, having more informational emojis might be deleterious to consumer engagement in the similar context. The findings suggest that social media content creators need to consider the textual conditions of their posts and emoji types to make their social conversation more engaging. We also find a negative and significant interaction effect between the number of emotional emojis and text divergence. This may be accounted by homeostasis theory in psychology (Cannon, 1929) that posits humans' tendency to maintain a state of psychological equilibrium by avoiding an excessive amount of stimulus or tension, introducing a caveat for content creators who intend to express their emotions excessively in social media posts using multiple methods (e.g., emojis, texts). In addition, we test how emojis are related to consumer engagement in two different types of social media posts: general posts vs. commercial posts. The analyses reveal interesting and counter-intuitive findings: having more emotional emojis has a positive effect on consumer engagement in commercial posts but a negative effect in general posts, and the negative effect of informational emojis could be worse for commercial posts than for general posts. These findings are opposed to the general practices of social media users, because social media users with commercial purposes usually focus on providing information about the products they sell and the motivations of self-expressions lead general users to participate in highly engaging activities, such as creating online content (e.g., De Vries et al., 2017; Muntinga et al., 2011). These findings align with prior studies showing that highly emotional contents or emotion-related content are positively associated with consumer engagement or viral behaviors in commercial contexts (Lee et al., 2018; Nikolinakou & King, 2018). We also find that there is a marginally significant relationship between time of a day and the presence of emojis, which suggests that the content creators who use emojis in their social media posts may want to publish them later in the day instead of early in order to maximize consumer engagement for their posts.

The study's key findings contribute to the existing literature in several ways. First, previous studies about emojis incorporate them as a binary variable (e.g., presence of emojis) and focus on their global impact on consumer reactions (Li et al., 2019). This study investigates both the impact of the presence or length of emojis and their effect at a more in-depth level. We categorize emojis into emotional and informational types, and examine their effects on consumer engagement. Second, extant papers have examined the conditional settings that explain the varied emoji effects, but they have focused on demographic, language, or cultural dimensions (Thollander & Kumar, 2019; Das et al., 2019). The current study focuses on more practical boundary conditions that social media post creators can actively adopt and utilize for their practices to produce more impactful content using emojis. Third, there is limited literature using real-world observational data for studying emojis. Most of the related studies employ survey research or experiments. We use observational data to study the impact of emojis on consumer engagement and their important contextual conditions that social media users may want to consider. Fourth, this work has implications for practitioners, notably content creators with strategic purposes, small businesses and firm managers seeking partnerships with general social media users to leverage UGC for marketing purposes. Few studies have provided practical guidelines about whether or how content creators can employ emojis when communicating with audiences. We believe that our findings regarding whether or not content creators should adopt emojis and what types of emojis are most effective in which contexts will add to our understanding of promotional communications with emojis in brand-related UGC. Finally, we consider that marketing communication managers may also take advantage of the findings about when it is beneficial or harmful to include emojis in commercial posts. We identify interesting counter-intuitive findings: including more emotional emojis has a positive effect on consumer engagement in commercial posts but a negative effect in general posts and using informational emojis is more harmful in commercial posts than in general posts. We believe that these results provide advertising managers with opportunities to devise more effective advertising content utilizing emojis.

In addition to the contributions discussed above, our study has several possible limitations, which may offer future research opportunities. First, we only study one social media platform, Instagram, although it is the third-ranked social network worldwide in terms of monthly active users, behind Facebook and YouTube (Statista, 2021). We expect that users may show different behaviors in using emojis in emerging social media platforms such as Snapchat or TikTok, which are used by younger people (e.g., Generation Z), or in messaging apps such as Facebook messenger or WhatsApp. On those platforms, the main usage of messengers is different from that of other social media: one-to-one interactions vs. one-to-many conversations. We hope that our paper encourages future research to involve not only other social media platforms but social messengers and investigate differences across platforms. Second, although we find interesting contextual settings where emojis can be more effectively incorporated into social media posts and account for the mechanisms underlying their occurrence using extant theories (e.g., homeostasis), we do not show the mechanism using empirical methods with data. Future research can investigate the mechanisms using survey data or controlled experiments, for example, to discover the reason why the amount of emotional emojis is likely to drive more consumer engagement in commercial posts than in general social media posts or to identify the underlying mechanism of the negative relationship between the amount of emotional emojis and consumer engagement in social media posts containing extreme emotions in the texts. Third, we focus on the text-related conditions and external factors that may affect how emojis influence consumer reactions. However, visual properties, such as visual sentiment (Ko & Bowman, 2018) or object types (Bakhshi et al., 2014), are increasingly essential in social conversations. Recent research (Shin et al., 2020; Tellis et al., 2019; Zhang & Luo, 2018) has revealed their primary roles in promoting consumers' positive reactions. Future research can examine the joint effect of images and emojis. Finally, emojis can be quantified in terms of sentiment (e.g., level of positivity or negativity) or other properties, and they can be further clustered based on their semanticity. We encourage future research studies to investigate more emoji properties and the contextual conditions in which the different types of emojis are effectively incorporated into social media posts.

Credit author statement

Eunhee (Emily) Ko: Idea generation, Conceptualization, Data Formal analysis, Methodology, Software, Validation, Writing – original draft preparation, reviewing and editing. Gosu Kim: Variable extractions, Writing – original draft preparation. Daewook Kim: Idea generation, Writing- Reviewing and Editing

Appendix. Robustness Checks

Table A.1
Robustness check 1 for Table 3: Trimming 1% and 5% of the data

	LIKES	*LIKES*	*COMMENTS*	*COMMENTS*
	Without 1%	Without 5%	Without 1%	Without 5%
	(1)	(2)	(3)	(4)
Emoji Variables				
HasEmoji	0.48***(.05)	0.46***(.05)	0.51***(.04)	0.50***(.04)
EmojiLength	−0.01***(.00)	−0.02***(.00)	−0.02***(.00)	−0.02***(.00)
EmojiPopHigh	0.38***(.06)	0.25***(.04)	0.37***(.02)	0.35***(.02)
EmojiPopMedium	0.03(.03)	0.01(.02)	0.02(.02)	0.01(.01)
EmojiPopLow	−0.06#(.04)	−0.053(.03)	−0.16***(.02)	−0.16***(.02)
Text Variables				
TextLength	0.01***(.00)	0.01***(.00)	0.02***(.00)	0.02***(.00)
TextLengthˆ2	−0.00004***(.00)	−0.00004***(.00)	−0.00006***(.00)	−0.00006***(.00)
TextVal	0.02***(.00)	0.02***(.00)	−0.003(.00)	−0.003(.00)
TextDiv	0.02***(.00)	0.02***(.00)	−0.04***(.00)	−0.04***(.00)
Control Variables				
Post-Level Controls:				
FaceCount	0.07***(.01)	0.07***(.01)	0.03***(.00)	0.03***(.00)
ImageQual	−0.002***(.00)	−0.001***(.00)	−0.001**(.00)	−0.001***(.00)
HasLiving	0.20***(.01)	0.20***(.01)	0.08***(.01)	0.08***(.01)
HasFood	−0.05***(.01)	−0.05***(.01)	−0.01#(.01)	−0.02*(.01)
HasPlant	0.14***(.02)	0.16***(.02)	0.02#(.01)	0.02*(.01)
PostAge (days)	−0.0004***(.00)	−0.0005***(.00)	0.0001**(.00)	0.0001**(.00)
TimeDay (hour)	0.004***(.00)	0.003***(.00)	0.001**(.00)	0.0008***(.00)
HasPunctuation	0.11***(.01)	0.08***(.01)	0.23***(.01)	0.21***(.01)
User-Level Controls:				
FollowerCount	0.00006***(.00)	0.0001***(.00)	0.00002***(.00)	0.00004***(.00)
FollowingCount	0.000009*(.00)	−0.00004***(.00)	−0.00002***(.00)	−0.00003***(.00)
PostCount	−0.00005***(.00)	−0.00006***(.00)	−0.00003***(.00)	−0.00002***(.00)
Brand Fixed-Effects	Yes	Yes	Yes	Yes
Pseudo R-Squared	.43	.41	.44	.43
AIC	84,861.34	72,749.31	61,662.88	54,387.32
N	30,989	28,459	30,989	28,459

#$p < .05$; *$p < .01$; **$p < .001$; ***$p < .0001$.

Table A.2
Robustness check1 for Table 5: Trimming 1% and 5% of the data

	LIKES	*LIKES*	*LIKES*	*LIKES*
	Without 1%	Without 5%	Without 1%	Without 5%
	(1)	(2)	(3)	(4)
Emoji Variables				
HasEmoji	0.46***(.06)	0.44***(.06)		
EmojiLength	−0.01***(.00)	−0.01***(.00)		
Emoji:Emo			0.01(.01)	0.004(.01)
Emoji:Info			−0.02**(.01)	−0.01*(.01)
EmojiPop:High	0.02(.04)	0.02(.05)	0.37***(.02)	0.30***(.02)
EmojiPop:Medium	−0.01(.03)	−0.02(.03)	0.06**(.02)	0.08***(.02)
EmojiPop:Low	−0.06(.04)	−0.04(.03)	−0.01(.03)	0.007(.03)
Text Variables				
TextLength	0.01***(.00)	0.01***(.00)	0.01***(.00)	0.01***(.00)
TextLengthˆ2	−0.00004***(.00)	−0.00003***(.00)	−0.00004***(.00)	−0.00004***(.00)
TextVal	0.02***(.00)	0.02***(.00)	0.02***(.00)	0.02***(.00)
TextDiv	0.02***(.00)	0.02***(.00)	0.02***(.02)	0.02***(.00)
Interaction Variables				
HasEmoji*HasPunctuation	−0.27***(.03)	−0.27***(.03)		
HasEmoji*TimeDay	0.0008(.00)	0.0008(.00)		
HasEmoji*GeneralPost	0.06*(.03)	0.07**(.03)		
Emoji:Emo*HasPunctuation			−0.001(.01)	−0.01(.01)
Emoji:Info*HasPunctuation			−0.009#(.00)	−0.01**(.00)
Emoji:Emo*TimeDay			0.000006(.00)	−0.001(.00)
Emoji:Info*TimeDay			0.00004(.00)	0.00009(.00)
Emoji:Emo*GeneralPost			−0.02#(.01)	−0.01#(.01)
Emoji:Info*GeneralPost			0.003(.00)	0.0007(.00)
Control Variables				
Post-Level Controls:				
FaceCount	0.06***(.01)	0.06***(.01)	0.06***(.01)	0.06***(.01)
ImageQual	−0.001**(.00)	−0.001**(.00)	−0.001**(.00)	−0.001***(.00)
HasLiving	0.18***(.01)	0.18***(.01)	0.18***(.01)	0.18***(.01)
HasFood	−0.05***(.01)	−0.05***(.01)	−0.06***(.01)	−0.05***(.00)

(continued on next page)

Table A.2 (*continued*)

	LIKES	*LIKES*	*LIKES*	*LIKES*
	Without 1%	Without 5%	Without 1%	Without 5%
	(1)	(2)	(3)	(4)
HasPlant	0.13***(.02)	0.15***(.02)	0.13***(.02)	0.15***(.02)
PostAge (days)	−0.0004***(.00)	−0.0005***(.00)	−0.0004***(.00)	−0.0005***(.00)
TimeDay (hour)	0.003***(.00)	0.003***(.00)	0.004***(.00)	0.003***(.00)
HasPunctuation	0.24***(.02)	0.20***(.02)	0.12***(.02)	0.10***(.02)
GeneralPost	0.32***(.02)	0.33***(.01)	0.34***(.01)	0.36***(.01)
User-Level Controls:				
FollowerCount	0.00006***(.00)	0.00002***(.00)	0.00006***(.00)	0.0002***(.00)
FollowingCount	0.00001***(.00)	0.00001***(.00)	0.00001***(.00)	0.000005#(.00)
PostCount	−0.00005***(.00)	−0.000006***(.00)	−0.00005***(.00)	−0.00006***(.00)
Brand Fixed-Effects	Yes	Yes	Yes	Yes
Pseudo R-Squared	.45	.43	.45	.43
AIC	84,123.87	71,869.27	84,348.80	72,096.85
N	30,989	28,459	30,989	28,459

#*p* < .05; **p* < .01; ***p* < .001; ****p* < .0001.

Table A.3
Robustness check2 Table 3: Removing text variables

	LIKES	*COMMENTS*
Emoji Variables		
HasEmoji	0.53***(.06)	0.54***(.04)
EmojiLength	−0.01***(.00)	−0.01***(.00)
EmojiPopHigh	0.35***(.04)	0.37***(.02)
EmojiPopMedium	0.02(.02)	0.02(.02)
EmojiPopLow	−0.19***(.04)	−0.20***(.03)
Text Variables		
TextLength	0.01***(.00)	0.02***(.00)
TextLength^2	−0.00005***(.00)	−0.00006***(.00)
Control Variables		
Post-Level Controls:		
FaceCount	0.07***(.01)	0.03***(.00)
ImageQual	−0.001*(.00)	−0.0007*(.00)
HasLiving	0.20***(.01)	0.08***(.01)
HasFood	−0.06***(.01)	−0.02#(.01)
HasPlant	0.15***(.02)	0.02#(.01)
PostAge (days)	−0.0004***(.00)	0.0001**(.00)
TimeDay (hour)	0.006***(.00)	0.002***(.00)
HasPunctuation	0.15***(.02)	0.25***(.01)
User-Level Controls:		
FollowerCount	0.00006***(.00)	0.000003***(.00)
FollowingCount	0.00004*(.00)	0.000009**(.00)
PostCount	−0.000009***(.00)	−0.00001***(.00)
Brand Fixed-Effects	Yes	Yes
Pseudo R-Squared	.32	.41
AIC	96,764.42	66,922.11
N	31,621	31,621

#*p* < .05; **p* < .01; ***p* < .001; ****p* < .0001.

Table A.4
Robustness check2 Table 3: Removing text variables

	LIKES	*COMMENTS*
	Without 1%	Without 1%
	(1)	(3)
Emoji Variables		
HasEmoji	0.52***(.07)	
EmojiLength	−0.01***(.00)	
Emoji:Emo		0.03(.02)
Emoji:Info		−0.02**(.01)
EmojiPop:High	0.03(.06)	0.48***(.02)
EmojiPop:Medium	−0.02(.04)	0.06*(.03)
EmojiPop:Low	−0.18***(.04)	−0.14**(.04)
Text Variables		
TextLength	0.01***(.00)	0.01***(.00)
TextLength^2	−0.00004***(.00)	−0.00005***(.00)
Interaction Variables		
HasEmoji*HasPunctuation	−0.29***(.03)	

(*continued on next page*)

Table A.4 (*continued*)

	LIKES	*COMMENTS*
	Without 1%	Without 1%
	(1)	(3)
HasEmoji*TimeDay	0.003#(.00)	
HasEmoji*GeneralPost	0.02(.03)	
Emoji:Emo*HasPunctuation		−0.003(.01)
Emoji:Info*HasPunctuation		−0.01*(.01)
Emoji:Emo*TimeDay		0.0004(.00)
Emoji:Info*TimeDay		0.0003(.00)
Emoji:Emo*GeneralPost		−0.05**(.02)
Emoji:Info*GeneralPost		0.01*(.01)
Control Variables		
Post-Level Controls:		
FaceCount	0.06***(.01)	0.06***(.01)
ImageQual	−0.001*(.00)	−0.001*(.00)
HasLiving	0.18***(.01)	0.18***(.01)
HasFood	−0.06***(.01)	−0.06***(.01)
HasPlant	0.14***(.02)	0.14***(.02)
PostAge (days)	−0.0004***(.00)	−0.0004***(.00)
TimeDay (hour)	0.005***(.00)	0.006***(.00)
HasPunctuation	0.29***(.02)	0.17***(.02)
GeneralPost	0.27***(.02)	0.28***(.02)
User-Level Controls:		
FollowerCount	0.000006***(.00)	0.000006***(.00)
FollowingCount	0.00005***(.00)	0.00004***(.00)
PostCount	−0.000007***(.00)	−0.000008***(.00)
Brand Fixed-Effects	Yes	Yes
Pseudo R-Squared	.33	.32
AIC	96,392.24	96,573.88
N	31,621	31,621

#$p < .05$; *$p < .01$; **$p < .001$; ***$p < .0001$.

References

Ai, W., Lu, X., Liu, X., Wang, N., Huang, G., & Mei, Q. (2017, April). Untangling emoji popularity through semantic embeddings. In *, Vol. 11. Proceedings of the international AAAI conference on web and social media*. No. 1.

Alboqami, H., Al-Karaghouli, W., Baeshen, Y., Erkan, I., Evans, C., & Ghoneim, A. (2015). Electronic word of mouth in social media: The common characteristics of retweeted and favorited marketer-generated content posted on twitter. *Journal of Internet Marketing and Advertising, 9*(4), 338–358.

Bakhshi, S., Shamma, D. A., & Gilbert, E. (2014, April). Faces engage us: Photos with faces attract more likes and comments on Instagram. In *Proceedings of the SIGCHI conference on human Factors in computing systems* (pp. 965–974).

Baltas, G. (2003). Determinants of internet advertising effectiveness: An empirical study. *International Journal of Market Research, 45*(4), 1–9.

Barbieri, F., Kruszewski, G., Ronzano, F., & Saggion, H. (2016, October). How cosmopolitan are emojis? Exploring emojis usage and meaning over different languages with distributional semantics. In *Proceedings of the 24th ACM international Conference on multimedia* (pp. 531–535).

Boutet, I., LeBlanc, M., Chamberland, J. A., & Colin, C. A. (2021). Emoji influence emotional communication, social attributions, and information processing. *Computers in Human Behavior, 119*, Article 106722.

Cannon, W. B. (1929). Organization for physiological homeostasis. *Physiological Reviews, 9*(3), 399–431.

Castillo, A., Benitez, J., Llorens, J., & Luo, X. (2021, June). Social media-driven customer engagement and movie performance: Theory and empirical evidence. *Decision Support Systems, 145*, 113516.

Chen, Z., Lu, X., Ai, W., Li, H., Mei, Q., & Liu, X. (2018, April). Through a gender lens: Learning usage patterns of emojis from large-scale android users. In *Proceedings of the 2018 world wide web conference* (pp. 763–772).

Chen, Q., Min, C., Zhang, W., Wang, G., Ma, X., & Evans, R. (2020). Unpacking the black box: How to promote citizen engagement through government social media during the COVID-19 crisis. *Computers in Human Behavior, 110*, Article 106380.

Cho, H., & Schwarz, N. (2006). If I don't understand it, it must be new: Processing fluency and perceived product innovativeness. *ACR North Amer. Advances, 33*(1), 319–320.

Coyle, M. A., & Carmichael, C. L. (2019). Perceived responsiveness in text messaging: The role of emoji use. *Computers in Human Behavior, 99*, 181–189.

Cramer, H., de Juan, P., & Tetreault, J. (2016, September). Sender-intended functions of emojis in US messaging. In *Proceedings of the 18th international Conference on human-computer Interaction with mobile Devices and services* (pp. 504–509).

Das, G., Wiener, H. J., & Kareklas, I. (2019). To emoji or not to emoji? Examining the influence of emoji on consumer reactions to advertising. *Journal of Business Research, 96*, 147–156.

De Vries, L., Gensler, S., & Leeflang, P. S. (2012). Popularity of brand posts on brand fan pages: An investigation of the effects of social media marketing. *Journal of Interactive Marketing, 26*(2), 83–91.

De Vries, L., Peluso, A. M., Romani, S., Leeflang, P. S., & Marcati, A. (2017). Explaining consumer brand-related activities on social media: An investigation of the different roles of self-expression and socializing motivations. *Computers in Human Behavior, 75*, 272–282.

Ding, C., Cheng, H. K., Duan, Y., & Jin, Y. (2017, February). The power of the "like" button: The impact of social media on box office. *Decision Support Systems, 94*, 77–84.

Goh, K. Y., Heng, C. S., & Lin, Z. (2013). Social media brand community and consumer behavior: Quantifying the relative impact of user-and marketer-generated content. *Information Systems Research, 24*(1), 88–107.

Gorn, G. J., Goldberg, M. E., & Basu, K. (1993). Mood, awareness, and product evaluation. *Journal of Consumer Psychology, 2*(3), 237–256.

Haberstroh, S. (2010). College counselors' use of informal language online: Student perceptions of expertness, trustworthiness, and attractiveness. *Cyberpsychology, Behavior and Social Networking, 13*(4), 455–459.

Hartmann, J., Heitmann, M., Schamp, C., & Netzer, O. (2021). *The power of brand selfies in consumer-generated brand imagery*. Columbia Business School Research Paper Forthcoming.

Hewage, G. S. U., Liu, Y., Wang, Z., & Mao, H. (2020). Consumer responses toward symmetric versus asymmetric facial expression emojis. *Marketing Letters*, 1–12.

Hill, J. H. (2016). *The impact of emojis and emoticons on online consumer reviews, perceived company response quality, brand relationship, and purchase intent*.

Hu, T., Guo, H., Sun, H., Nguyen, T. V., & Luo, J. (2017, May). Spice up your chat: The intentions and sentiment effects of using emojis. In *, Vol. 11. Proceedings of the international AAAI conference on web and social media*. No. 1.

Jones, L. L., Wurm, L. H., Norville, G. A., & Mullins, K. L. (2020). Sex differences in emoji use, familiarity, and valence. *Computers in Human Behavior, 108*, Article 106305.

Kaye, L. K., Rodriguez-Cuadrado, S., Malone, S. A., Wall, H. J., Gaunt, E., Mulvey, A. L., & Graham, C. (2021). How emotional are emoji? Exploring the effect of emotional valence on the processing of emoji stimuli. *Computers in Human Behavior, 116*, Article 106648.

Ko, E., & Bowman, D. (2018). *Content engineering of images: The effect of sentiment and complexity on consumer engagement with brand-themed user-generated content*. Working Paper.

Lee, D., Hosanagar, K., & Nair, H. S. (2018). Advertising content and consumer engagement on social media: Evidence from Facebook. *Management Science, 64*(11), 5105–5131.

Li, X., Chan, K. W., & Kim, S. (2019). Service with emoticons: How customers interpret employee use of emoticons in online service encounters. *Journal of Consumer Research, 45*(5), 973–987.

Lovett, M., Peres, R., & Shachar, R. (2014). A data set of brands and their characteristics. *Marketing Science, 33*(4), 609–617.

Lu, X., Ai, W., Liu, X., Li, Q., Wang, N., Huang, G., & Mei, Q. (2016, September). Learning from the ubiquitous language: An empirical analysis of emoji usage of smartphone users. In *Proceedings of the 2016 ACM international joint Conference on Pervasive and ubiquitous computing* (pp. 770–780).

Molina, A. M., Rojas-de Gracia, M. M., Alarcón-Urbistondo, P., & Romero-Charneco, M. (2019). Exploring the opportunities of the emojis in brand communication: The case of the beer industry. *Inter. J Business ComInfor Sys Res, 2329488419832964*.

Muntinga, D. G., Moorman, M., & Smit, E. G. (2011). Introducing COBRAs: Exploring motivations for brand-related social media use. *Inter J Adv, 30*(1), 13–46.

Nikolinakou, A., & King, K. W. (2018). Viral video ads: Emotional triggers and social media virality. *Psy Marketing, 35*(10), 715–726.

Riordan, M. A. (2017). The communicative role of non-face emojis: Affect and disambiguation. *Computers in Human Behavior, 76*, 75–86.

Schwarz, N., & Clore, G. L. (1983). Mood, misattribution, and judgments of well-being: Informative and directive functions of affective states. *J Personality Soc Psy, 45*(3), 513–523.

Shin, D., He, S., Lee, G. M., Whinston, A. B., Cetintas, S., & Lee, K. C. (2020). Enhancing social media analysis with visual data analytics: A deep learning approach. *MIS Quarterly*, 1459–1492.

Statista. (2021). *Most popular social networks worldwide as of July 2021, ranked by number of active users*. statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/ . (Accessed 13 August 2021).

Stieglitz, S., & Dang-Xuan, L. (2013). Emotions and information diffusion in social media—sentiment of microblogs and sharing behavior. *Journal of Management Information Systems, 29*(4), 217–248.

Tellis, G. J., MacInnis, D. J., Tirunillai, S., & Zhang, Y. (2019). What drives virality (sharing) of online digital content? The critical role of information, emotion, and brand prominence. *Journal of Marketing, 83*(4), 1–20.

Thelwall, M., Buckley, K., Paltoglou, G., Cai, D., & Kappas, A. (2010). Sentiment strength detection in short informal text. *Journal of the American Society for Information Science and Technology, 61*(12), 2544–2558.

Thollander, P., & Kumar, N. (2019, May). Examining the "global" language of emojis: Designing for cultural representation. In *Proceedings of the 2019 CHI Conference on human Factors in computing systems* (pp. 1–14).

Toubia, O., & Stephen, A. T. (2013). Intrinsic vs. image-related utility in social media: Why do people contribute content to twitter? *Marketing Science, 32*(3), 368–392.

Ye, Q., Law, R., Gu, B., & Chen, W. (2011). The influence of user-generated content on traveler behavior: An empirical investigation on the effects of e-word-of-mouth to hotel online bookings. *Computers in Human Behavior, 27*(2), 634–639.

Zeng, X., & Wei, L. (2013). Social ties and user content generation: Evidence from Flickr. *Information Systems Research, 24*(1), 71–87.

Zhang, S., Lee, K., Singh, P., & Srinivasan, K. (2021). What makes a good image? Airbnb demand analytics leveraging interpretable image features. *Management Science*. https://doi.org/10.1287/mnsc.2021.4175

Zhang, M., & Luo, L. (2018). *Can user generated content predict restaurant survival: Deep learning of yelp photos and reviews*. Social Science Research Network (SSRN).